Weak divisibility and shared sign certificates for Diophantine consistency

A research contribution prepared for Vladimir Reshetnikov's ProveIt project

With AI assistance; ordinary proofs and reproducible arithmetic artifacts

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Abstract

We give new explicit integer polynomials whose lack of integer solutions is, using the repository's stated ordinary calibration theorems, provably equivalent in ZFC to Con(ZFC). The principal small-variable equation has eleven unrestricted integer unknowns, total degree 416, and 129 live binary arithmetic operations. A companion has degree 464 and 123 operations. The closest inspected incoming eleven-integer equation has degree 688 and 126 operations. We also give a direct quartic in 478 unrestricted integers, built from the numerical proof calculus C222 without K215, Jones's reduction, or a large fixed coefficient tuple. Its 305 quadratic residuals expand to 5,484 monomials, with maximum absolute coefficient 511,326.

The first improvement combines a weak fifth-power congruence, a mixed modulus carrier, coprime fusion, and interpolation with a fixed strong Pell witness. Three-adic root separation removes an unnecessary even multiplier and permits a cheaper center. The direct improvement proves some source values positive earlier, allowing pairs of sign conditions to share one Diophantine witness. A joint guard also handles the signs and nonvanishing needed by the shared Pell frame. Full coefficient files, literal polynomial graphs, deterministic builders, and independent finite checks accompany the ordinary proofs. The work does not claim global minimality, a proof of consistency, or a new end-to-end proof-assistant derivation.

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1 Results, comparison, and logical scope

The problem is to specify a fixed polynomial $P \in \mathbb{Z}[x_1, \ldots, x_m]$ and prove

$$\mathsf{ZFC} \vdash (\forall \mathbf{x} \in \mathbb{Z}^m \; P(\mathbf{x}) \neq 0) \longleftrightarrow \mathrm{Con}(\mathsf{ZFC}). \tag{1}$$

All variables in this paper are unrestricted integers unless explicitly stated otherwise. A polynomial abbreviation is a shared subexpression, not an extra existentially quantified coordinate. Fixed numeral ports are actual supplied integers, not free parameters.

There is no single agreed ordering of simplicity: variable count, degree, arithmetic length, coefficient height, and the complexity of the calibration can favor different equations. We give measured improvements on two existing routes. We do not give a global lower bound on possible constructions. In particular, ten-integer constructions already occur in the repository, and a recent general ten-unknown reduction is available in the literature[4]. An eleven-variable degree improvement is not a new minimum-variable theorem.

Eleven-integer construction	Variables	Degree	Operations
Incoming coprime fusion, centered	11	688	126
Incoming coprime fusion, direct	11	728	122
Incoming all-radicand fusion	11	1632	114
New cubic chart, direct	11	488	120
New cubic chart, centered	11	464	123
New mixed chart, direct	11	456	126
New mixed chart, centered	**11**	**416**	**129**

Table 1: All four new equations use the same complete six-port tuple. Operations count every live binary $+, -, \times$, including scalar products and squares; leaves and reuse are free.

Degree 416 is 39.53% below 688, at three additional operations. The 464/123 companion improves both degree and arithmetic length over the 688/126 predecessor. The 456/126 companion keeps that predecessor's operation count and lowers its degree by 33.72%.

Direct integer quartic	Variables	Rows	Terms	Height
Matched incoming derived-label form	517	310	6774	3904267
New shared-sign form	**478**	**305**	**5484**	**511326**

Table 2: Rows are quadratic residuals, terms are collected monomials of their sum of squares, and height is the largest absolute coefficient.

The direct result improves all four displayed measures of its matched predecessor. Other incoming quartics trade more variables for smaller coefficients; this table is not a claim of dominance over every known quartic on every measure.

Source snapshot. The comparison snapshot is ProveIt commit `45b399ec51df`; the complete identifier is recorded in `SOURCE_SNAPSHOT.json`. We inspected the current research comparison, the numbered collection, and all eight incoming archives at that revision. The immediate ingredients are the incoming *Coprime Projection*, *Powered Saturation and Factorial Carriers*, *Carriers and Even Indices*, and *Quadratic Frames* packages[1, 2]. The first three already contain the individual weak-congruence, mixed-carrier, and fusion ideas used here. Our claim concerns their new compatible composition, its smaller literal equations, and the direct sign reductions proved below. We make no literature-wide priority claim for the component tricks.

What is proved and what is inherited. There are three distinct levels of evidence.

1. The new arithmetic reductions are proved by ordinary mathematics, including all zero branches, negative signs, and the order of existential witness selection.
2. The literal files are checked by exact finite algebra, authenticated fixed numerals, exact expansion or degree certificates, and deterministic reconstruction. These checks do not prove the unbounded mathematical theorems.
3. The consistency interpretation uses the named ordinary source calibration. The small-variable route uses the native compiler and the K215 endpoint. The direct route uses C222 refutability, its ZF calibration, and the ZF–ZFC refutation equivalence. Inherited Lean endpoint texts identify these interfaces, but no full Lean/Rocq dependency closure is rebuilt in this release.

The unconditional new reduction can therefore also be read independently as a zero-existence equivalence with the cited predecessor arithmetic source. Composing with that source's proved ordinary calibration gives (1). The proof uses no assumption that ZFC is consistent.

2 The complete eleven-integer equation

Fix the six integers

$$\beta, K_0, M_C, M_F, A_0, B_0$$

specified in Section 8. In particular

$$\beta = 2^{1064921} - 1, \qquad A_0 = 2^{1113}, \qquad B_0 = 2^{666104}.$$

The high mask M_F is already shifted. No coefficient is selected by waiting for a proof or a computation to halt.

The eleven independent variables are

$$(j, C, z, h, r, v, w, T, c, n, s) \in \mathbb{Z}^{11}. \tag{2}$$

For the principal *mixed chart*, define

$$\begin{aligned} J &= 6j + 1, & q &= \beta J + 1, & W &= 12w + 4, \\ X &= qW, & Y &= qv, & E &= XY, \\ a &= E + Y, & H &= 4a + 3, & \Delta &= a^2 + H = (a+2)^2 - 1. \end{aligned} \tag{3}$$

Define the packing and budget polynomials

$$\begin{aligned} F &= (K_0 + W)C - z(q-1), \\ R &= \big(q(q-F) - C\big)(q^2-1) + (M_C + qM_F)J, \\ k &= R + 1 + hE, \qquad \eta = c - Yk, \\ E_0 &= (A_0F - q)^2 + (B_0C - q)^2, \\ B &= k^2(q^2 - E_0) - 4q^2\eta^2. \end{aligned} \tag{4}$$

The two fused conditions and the strong and auxiliary conditions are

$$\begin{aligned} \ell &= XY^2, & L &= \ell k, & D_t &= 1 + L(L+k), \\ D_b &= 3B - 4, & D_g &= 1 + D_bD_ts^2, & P &= D_tD_g, \\ I &= rc^2, & D_f &= 1 + \Delta I^2, & Q &= \Delta^2 I^2, \\ \delta &= Q - 1, & V &= D_f(c - R + 4IT) - c, & & \\ C_y &= QV^2 - 1, & D_a &= \delta C_y. & & \end{aligned} \tag{5}$$

The common scalar projection and mixed carrier are

$$\begin{aligned} K &= X(3c-2X)+2, & M &= 2q^2, & D &= MH,\\ C_* &= MK + H(v-q^2), & z_* &= DT + C_*, & & \\ A_* &= (z_*^2+3D^2)(z_*^3-19D^3), & S &= srM^5H. & & \end{aligned} \tag{6}$$

Finally put

$$\begin{aligned} Z &= S(n+3QV) - A_*,\\ u &= S^2D_f, & b &= 9S^2D_a, & e &= 81S^2P. \end{aligned} \tag{7}$$

The promised single equation is

$$\boxed{\Big[(Z^2-u-b-e)^2-4(ub+ue+be)\Big]^2 - 64ubeZ^2 = 0.} \tag{8}$$

Every displayed symbol is a polynomial abbreviation in (2). There are no variable exponents, square roots, divisions, bounded quantifiers, or hidden witnesses in this definition. The expression has integer coefficients and is evaluated by the supplied 129-operation graph.

Three companion equations. Dropping $3QV$ from the definition of Z gives the degree-456 direct mixed chart. It is related to the principal form by an integral translation of n, since QV does not depend on n. For the *cubic chart*, replace $Y = qv$ by

$$Y = (2v+1)q^3,$$

and replace the carrier definitions by

$$D = H, \qquad z_* = HT + K, \qquad A_* = (z_*^2+3H^2)(z_*^3-19H^3), \qquad S = srH. \tag{9}$$

Keep the other definitions unchanged. With center $3QV$ the degree is 464; without it the degree is 488.

3 Three elementary mechanisms

3.1 An exact norm with only three roots

Define the integer polynomial

$$\mathcal{N}_3(Z;u,b,e) = \Big[(Z^2-u-b-e)^2-4(ub+ue+be)\Big]^2 - 64ubeZ^2. \tag{10}$$

In a formal algebra with square roots adjoined,

$$\mathcal{N}_3(Z;u,b,e) = \prod_{\epsilon\in\{\pm1\}^3} \left(Z - \epsilon_1\sqrt{u} - \epsilon_2\sqrt{b} - \epsilon_3\sqrt{e}\right). \tag{11}$$

Independent sign invariance removes odd radical powers. The identity is valid for zero, negative, or dependent radicands as well as for positive ones. It has 35 collected monomials, all checked exactly in the release. A zero of (8) therefore makes at least one of the eight factors vanish over $\mathbb{C}$.

Lemma 3.1 (Three-adic separation)**.** *Let A, B, C be positive integers with $A \equiv B \equiv 1 \pmod 3$. If a signed sum*

$$\epsilon_1\sqrt{A} + 3\epsilon_2\sqrt{B} + 9\epsilon_3\sqrt{C}$$

is rational, all three roots are integers.

Proof. Write each radicand as a square times a positive squarefree integer. Distinct squarefree radicals are linearly independent over $\mathbb{Q}$. The first two squarefree parts are not divisible by 3, and their integer coefficients have respective 3-adic orders 0 and 1. If the third squarefree part is divisible by 3, its irrational term cannot join either of them. Otherwise its coefficient has order at least 2. Within every nontrivial squareclass, the term of least 3-adic order is unique and cannot cancel. No such class can occur. Rational square roots of integers are integers. □

This separation lets us use $I = rc^2$. The preceding parity separation used an even multiplier $2src^2$. We need neither its factor 2 nor its explicit factor s in the new strong ordinate. Completeness will choose I divisible by a prescribed multiple of s without making that factor part of the polynomial.

3.2 Coprime fusion of the first norm and the guard

For any integers in the displayed construction,

$$\gcd(D_t, D_g) = \gcd(D_t, 1 + D_b D_t s^2) = 1. \tag{12}$$

Thus if D_t, D_g are positive and their product P is a square, each factor is a square. This is the inherited selective fusion idea; we use it together with the weaker carrier.

There is also a small literal saving in the guard. Since q, A_0, B_0 are even, E_0 and B are divisible by 4. Hence

$$D_b = 3B - 4 \equiv 0 \pmod 4, \qquad D_g \equiv 1 \pmod 4. \tag{13}$$

The predecessor's additional factor 4 in $D_g - 1$ is unnecessary. This conclusion uses actual coefficient parity; it is not a statement for arbitrary replacements of the six fixed ports.

3.3 A fifth-power congruence is enough

Write

$$\pi(t) = (t^2 + 3)(t^3 - 19), \qquad \Phi_D(z) = D^5 \pi(z/D) = z^5 + 3D^2 z^3 - 19D^3 z^2 - 57D^5. \tag{14}$$

The familiar intersective polynomial π has a root modulo every positive integer and no rational root[3]. For completeness, when $p \neq 2, 3, 19$ either $p \equiv 2 \pmod 3$ and cubing is bijective, or $p \equiv 1 \pmod 3$ and -3 is a quadratic residue. Selected roots are simple and lift. At $p = 19$ the quadratic has a simple root; at $p = 2$ the cubic does. At $p = 3$, write $t = 1 + 3u$; the equation $(1 + 3u)^3 = 19$ reduces to

$$u + 3u^2 + 3u^3 = 2,$$

which has a simple root modulo 3 and lifts to all powers. CRT finishes. The quadratic factor has no real zero and 19 is not a rational cube, proving the no-rational-root assertion.

Lemma 3.2 (Strong and weak parts of one carrier)**.** *Let M, H be nonzero coprime integers, let $D = MH$, and let*

$$z = DT + MK + H\omega.$$

If $M^5 H \mid \Phi_D(z)$, then

$$M \mid \omega, \qquad H \mid K^5.$$

Proof. For a prime with $p^e \mid M$ and $v_p(M) = e$, suppose $v_p(z) = b < e$. The leading term z^5 has order $5b$, while each other term in (14) has strictly greater order. Thus M^5 cannot divide $\Phi_D(z)$, a contradiction. Consequently $M \mid z$, and coprimality gives $M \mid \omega$. Reducing modulo H gives $H \mid (MK)^5$; cancel the unit M modulo H. □

Lemma 3.3 (Height resolves weak divisibility)**.** *Suppose $x \geq 2^{64}$, $t \geq 24$, and*

$$|H| > x(x-3)^t, \qquad |\Pi| < 2^{4t+4}x^2.$$

If $H \mid \Pi^5$, then $\Pi = 0$.

Proof. Since $x - 3 > x/2$,

$$|H| > \frac{x^{t+1}}{2^t}, \qquad |\Pi|^5 < 2^{20t+20}x^{10}.$$

The sufficient inequality is $x^{t-9} > 2^{21t+20}$. Its base-two exponent margin is at least

$$64(t-9) - (21t+20) = 43t - 596 \geq 436.$$

Thus $|\Pi|^5 < |H|$. The only multiple of H of that magnitude is zero. □

The new scale spends the very large actual coefficient bound on this weaker congruence. It never makes the false inference $H \mid K^5 \Rightarrow H \mid K$ for arbitrary integers.

4 Soundness of the eleven-integer construction

We prove the mixed chart first. The cubic chart omits the recovery of its scale, since its scale is already present in the definition.

4.1 Uniform facts and the zero branches

The fixed β is odd and $\beta \equiv 1 \pmod 3$. Thus q is even, $q \equiv 2 \pmod 3$, and $|q| \geq \beta - 1$. Also

$$W \equiv 1 \pmod 3, \quad X \equiv 2 \pmod 3, \quad |X| \geq 4(\beta - 1) > 2^{64}.$$

If $v = 0$, then $a = 0$, $H = 3$, and $\Delta = 3$. Otherwise $|a|$ is enormous. In every case H is nonzero and odd, $\Delta > 1$, and

$$a \equiv H \equiv \Delta \equiv 0 \pmod 3.$$

Because $Y = qv$, $H \equiv 3 \pmod q$. As $3 \nmid q$,

$$\gcd(M, H) = 1, \qquad M = 2q^2. \tag{15}$$

We have $D_f \geq 1$ and $D_t = 1 + \ell(\ell+1)k^2 \geq 1$. Also Q is either 0 or greater than 1, so δ never vanishes.

Suppose (8) holds. If $S = 0$, all scaled radicands vanish and the equation becomes $A_*^8 = 0$. But

$$A_* = D^5\pi(T + C_*/D) \neq 0$$

by the no-rational-root property and $D \neq 0$. Hence $S \neq 0$, and in particular $s, r \neq 0$.

We do *not* yet assume $c \neq 0$. If $c = 0$, then $I = Q = 0$ and $D_a = 1$. If $c \neq 0$, then $|I| \geq c^2$ and $D_f > |c| + 1$. Writing $b_0 = c - R + 4IT$, either $b_0 = 0$ and $|V| = |c| \geq 1$, or $|V| \geq D_f - |c| > 1$. Therefore $D_a > 0$ in both cases. The roots associated with D_f and D_a are real.

If $D_g < 0$, the $P = D_tD_g$ root would be the sole nonzero imaginary term in a vanishing conjugate of (11), impossible. Equation (13) excludes $D_g = 0$. Thus $D_g > 0$. Since B is a multiple of 4, $B \leq 0$ would imply $D_b \leq -4$ and $D_g \leq -3$. We conclude

$$B \geq 4, \quad k \neq 0, \quad E_0 < q^2, \quad \left|\frac{c}{k} - Y\right| < \frac{1}{2}. \tag{16}$$

4.2 Extract roots before using the restored scale

Modulo 3 we have $D_f \equiv D_a \equiv 1$. Divide a vanishing conjugate by S. Its radical sum is

$$\epsilon_f\sqrt{D_f} + 3\epsilon_a\sqrt{D_a} + 9\epsilon_p\sqrt{D_tD_g} \in \mathbb{Q}.$$

Lemma 3.1 makes all three roots integral. Equation (12) then yields integers $f, \tau, g, \widetilde{y}$ with

$$f^2 = D_f, \qquad \tau^2 = D_t, \qquad g^2 = D_g, \qquad \widetilde{y}^2 = D_a. \tag{17}$$

The conjugate factor also yields $S \mid A_*$. No adjacent-Pell descent or nonzero strong ordinate has been used to obtain this divisibility.

Apply Lemma 3.2 with $\omega = v - q^2$. It gives

$$v = q^2(2v_0 + 1) \quad (v_0 \in \mathbb{Z}), \qquad H \mid K^5. \tag{18}$$

Thus $Y = q^3(2v_0 + 1)$, the exact old cubic scale. In particular $Y \neq 0$. Now if $c = 0$, the budget is

$$B = k^2\left(q^2(1 - 4Y^2) - E_0\right) \leq 0,$$

contradicting (16). Hence $c, I \neq 0$ and $Q > 1$. This ordering explicitly handles the initially possible branch $v = c = 0$.

4.3 The auxiliary root is integral

Put $y = \widetilde{y}/\delta \in \mathbb{Q}$. Then

$$QV^2 - (Q - 1)y^2 = 1. \tag{19}$$

Here $V \in \mathbb{Z}$ and $Q > 1$. We recall the elementary descent rather than silently treating a rational root as integral. In lowest terms $y = a/b$, $b^2 \mid Q - 1$. For positive $V > 1$ and $y > 0$, use

$$V' = (2Q - 1)V - 2(Q - 1)y, \qquad y' = (2Q - 1)y - 2QV.$$

The norm is preserved, V' is integral, and the reduced denominator of y' remains b, since $2Q - 1 \equiv 1 \pmod{b}$. The norm gives $V < y < QV/(Q - 1)$, hence $-V < V' < V$; $V' = 0$ is impossible. Taking absolute values and iterating reaches $|V| = 1$, where $|y| = 1$. Thus $b = 1$. We have recovered all three ordinary Pell conditions.

4.4 The inherited rank interface, in the necessary order

The budget disk implies the common strict signs of $q, F, C, q - F - C$:

$$0 < \frac{A_0F}{q} < 2, \qquad 0 < \frac{B_0C}{q} < 2, \qquad 0 < \frac{F}{q} + \frac{C}{q} < 1.$$

The following inherited ordinary interface is the precise input from the centered signed source[2].

Proposition 4.1 (Preprojection rank interface, inherited)**.** *Assume the authentic six coefficient ports, the packing equations, $Y = q^3(2v_0 + 1)$, the common strict signs above, the centered strip $|c/k - Y| < 1/2$, $I \neq 0$, $c^2 \mid I$, $|I| \geq c^2$, and (17)–(19). Then, before using any main projection congruence,*

$$R = 2t + 1, \quad t \geq 24, \quad c = \sigma\psi_{a+2}(R), \quad |Y| > \frac{(|X| - 3)^t}{2} - 1, \qquad \sigma = \operatorname{sgn}(c).$$

Here ψ_A is the Pell polynomial sequence with $\psi_A(0) = 0$, $\psi_A(1) = 1$, and $\psi_A(m + 2) = 2A\psi_A(m + 1) - \psi_A(m)$. No parity assumption on I/c^2 is required.

For clarity, its main steps are as follows. Absolute Pell folding first recovers $|c| = \psi_{|a+2|}(p)$ using $V \equiv -c \pmod f$, the strong norm, and the auxiliary norm. The inequality $2|I| < |f|$ prevents wrapping. The first norm supplies $|k| = 2\psi_{|2XY^2+1|}(m)$, and the packing bound places its index in the required large range. Strong divisibility gives $p \mid m_0$ for the strong index m_0. Writing $m_0 = pv$ and using $c^2 \mid I$ in the Lucas multiplication formula gives $|c| \mid v$, so $|c| \mid m_0$. Finally

$$V + R = \Delta I^2(c - R) + 4ID_fT$$

implies $V \equiv -R \pmod c$. Signed index reduction and the strict index bounds identify $p = R$. Pell growth supplies the displayed ratio. The copied ordinary proofs contain the complete index step-down and uniform estimates; they are not replaced by finite test cases here.

Let $x = |X|$ and $m = |Y|$. The ratio bound gives

$$|H| \geq 4(x-1)m - 3 > x(x-3)^t.$$

The last inequality follows since $(x-2)(x-3)^t - 4x + 1 > 0$ for $x \geq 64$, $t \geq 1$. Set

$$K_q = X(X + 2ac) - 1, \qquad \Pi = (\sigma X - 2^R)(2^R \sigma X + 1). \tag{20}$$

The exact algebraic identity and inherited signed Pell congruence are

$$K = HXc - 2K_q, \qquad 2^R K_q \equiv \Pi \pmod H. \tag{21}$$

Since H is odd, $H \mid K^5$ implies $H \mid \Pi^5$. Both factors of Π have magnitude below $2^{R+1}x$, so $|\Pi| < 2^{4t+4}x^2$. Lemma 3.3 yields $\Pi = 0$. The second factor is odd and nonzero; therefore

$$\sigma X = 2^R. \tag{22}$$

The inherited centered half-binomial argument now restores positive orientation and $0 < \eta/k < 1/4$. One can regenerate fresh odd strong and auxiliary witnesses for the exact old signed source, keeping the outer coordinates fixed. Its ordinary native/compiler calibration then supplies a finite syntactic ZFC refutation. This completes soundness without assuming the missing main norm during the earlier root extraction or rank recovery.

5 Completeness with a fixed strong witness

Start with a finite syntactic ZFC refutation and the canonical outer data from the same ordinary source theorem. It supplies the cubic scale $Y = q^3(2v_0 + 1)$, the correct packing equations, and

$$E_0 \leq q^2/2, \qquad 0 < \eta/k < 1/4, \qquad B > q^2k^2/4 \geq 64, \qquad D_t = \tau^2 > 0.$$

It also supplies the even Pell base $A = a + 2 > 1$, $R \equiv 3 \pmod 4$, and $c = \psi_A(R) \equiv R \pmod 4$, as proved in Section 7 of the retained `centered11-proof.md`. These are the interpolation hypotheses used below. For the mixed chart set $v = q^2(2v_0 + 1)$.

Unconditionally $D_t \equiv 1 \pmod 3$: $\ell = XY^2$ is either 0 or 2 modulo 3. Hence $(3B - 4)D_t$ is positive and $2 \pmod 3$, so it is not a square. Pell existence supplies $s, g > 0$ with

$$g^2 - (3B - 4)D_t s^2 = 1.$$

This is exactly $D_g = g^2$; the former squaring of a Pell unit is unnecessary after deleting the extra factor 4.

5.1 Supported-modulus interpolation

We use the following strengthening proved in the incoming powered saturation package. We include its mechanism because it is the reason the factor s can be removed from I.

Theorem 5.1 (Supported-modulus interpolation, inherited)**.** *For the canonical even Pell base $A = a + 2 > 1$, index $R \equiv 3 \pmod 4$, and $c = \psi_A(R) > 0$ with $c \equiv R \pmod 4$, choose one positive even strong index m_0 with*

$$I = \psi_A(m_0), \quad f = \chi_A(m_0), \quad 4sc^2 \mid I.$$

For this same fixed pair I, f, every residue class modulo any positive integer whose prime factors divide I contains a T for which

$$V = f^2(c - R + 4IT) - c, \qquad (\Delta I)^2 V^2 - ((\Delta I)^2 - 1)y^2 = 1$$

has an integer y.

Here χ_A is the companion Pell sequence. A finite matrix period provides the prescribed strong multiple. To see why the modulus can have larger prime powers than I, set

$$\rho = \Delta I, \quad \lambda = 4m_0 f, \quad d = \operatorname{lcm}(\lambda, 4I).$$

The squared-modulus period is λ, not merely $4m_0$. For $p \mid I$, the Lucas valuation inequality $v_p(m_0) \le v_p(I)$ and $\gcd(f, I) = 1$ imply

$$v_p(d) = v_p(4I).$$

Start from one admissible odd auxiliary index $\nu_0 \equiv 3 \pmod 4$ and vary it through $\nu_0 + dh$ for $h \ge 0$. With $V_\nu = \chi_\rho(\nu)/\rho$, put

$$T(h) = \frac{V_{\nu_0 + dh} + c - f^2(c - R)}{4If^2}.$$

These are integers. Pell addition and elementary lifting give the exact isometry

$$v_p(T(h_1) - T(h_2)) = v_p(h_1 - h_2) \qquad (p \mid I,\ h_1, h_2 \ge 0,\ h_1 \ne h_2). \tag{23}$$

For example, for d_0 divisible by 4 and $p \mid \rho$,

$$v_p\left(\psi_\rho(d_0)/\rho\right) = v_p(d_0),$$

while the other summand in the Pell addition formula has strictly higher valuation. The formulas follow by expanding powers of the fourth unit, whose ordinate is $4\rho(2\rho^2 - 1)$; the doubling formula handles $p = 2$. Dividing by $4If^2$ proves (23). It makes the map bijective modulo every p^e with $p \mid I$, and CRT proves the theorem. The full proof, including the two-adic valuation and the base auxiliary index, is retained as `inherited/norm/powered_saturation_proof.tex`.

5.2 Finish the witness construction

Choose the strong pair once as in Theorem 5.1 and set

$$r = I/c^2 > 0.$$

Every prime factor of sr divides I. In the mixed chart the canonical projection and scale give

$$\kappa = \frac{K}{H} + \frac{v - q^2}{M} \in \mathbb{Z}, \qquad A_* = (MH)^5 \pi(T + \kappa).$$

Intersectivity supplies a residue a_0 modulo sr for which $\pi(a_0+\kappa)\equiv 0 \pmod{sr}$. Only now request the auxiliary coordinate $T\equiv a_0 \pmod{sr}$ from the already fixed strong pair. It follows that

$$\frac{A_*}{S}=\frac{H^4\pi(T+\kappa)}{sr}\in\mathbb{Z}.$$

Choose

$$n=A_*/S+f+3\delta y+9\tau g-3QV. \tag{24}$$

One conjugate factor is zero, proving completeness. The cubic chart uses $\kappa=K/H$ and the identical argument. The direct companions omit the last term in (24).

The witness order is important: outer native data, then s,g, then one fixed strong pair I,f,r, then an intersective residue, then T,y, and finally n. No modulus is repaired by changing I after a residue has been chosen.

Theorem 5.2 (Eleven-integer consistency equations)**.** *Each of the four supplied eleven-integer polynomials has an integer zero if and only if the pinned ordinary source has a finite ZFC refutation. Composed with that source's ordinary calibration, ZFC proves* (1) *for each polynomial. Their exact degrees and arithmetic lengths are those in Table 1.*

Proof. The two existence directions were established in Sections 4 and 5. All new arguments are elementary arithmetic, Pell identities, finite group periods, and CRT, formalizable in ZFC. The next section proves exact degree, and the finite circuit audit recounts arithmetic length. Negating the finite-refutation existence statement yields the claimed equivalence. This does not establish either side of that equivalence. □

6 Exact degree, cancellation, and arithmetic length

The degree reductions are polynomial identities, not degree estimates from evaluations at random points. Table 3 records the relevant exact degrees; $B_*=Sn-A_*$ and $C_0=3SQV$.

Chart	S	D_f	Q	V	D_a	P	A_*	C_0
Cubic	8	18	30	23	106	94	35	61
Mixed	16	14	22	19	82	62	35	57

Table 3: The mixed budget has degree 16, not 14: its $-4q^2\eta^2$ term dominates. This affects P and the final degree.

The exact identity

$$b-C_0^2=9S^2(-Q(V^2+1)+1) \tag{25}$$

shows that $\omega=Z^2-b$ has degree at most 96 on the cubic chart and at most 92 on the mixed chart. The full three-root norm can be rewritten without changing the polynomial as

$$\Big[(\omega-u-e)^2-4\{ue+b(u+e)\}\Big]^2-64ubeZ^2. \tag{26}$$

For the mixed chart, $\deg(u,b,e)=(46,114,94)$. The unique highest part of the inner bracket is $-4b_0e_0$, of degree 208; the outer correction has degree at most 368. Thus the exact leading form is $16b_0^2e_0^2$ and the degree is 416. For the cubic chart, $\deg(u,b,e)=(34,122,110)$, giving degree 464; its outer correction has degree at most 388.

For an explicit nonvanishing certificate at the actual fixed tuple, the mixed polynomial has

$$[j^{216}w^{52}v^{64}h^{12}z^4s^{12}r^{24}c^{32}]P_{416}=2^{384}3^{282}\beta^{216}>0. \tag{27}$$

The cubic centered polynomial has

$$[j^{264}w^{52}v^{64}h^{12}z^4s^{12}r^{24}c^{32}]P_{464} = 2^{456}3^{330}\beta^{264} > 0. \tag{28}$$

The exponent sums are respectively 416 and 464.

One way to verify the mixed certificate is to put

$$q_0 = 6\beta j, \quad X_0 = 12q_0w, \quad Y_0 = q_0v, \quad R_0 = -q_0^3(12wC - q_0z), \quad k_0 = R_0 + hX_0Y_0.$$

The highest homogeneous form is

$$2^{64}3^{14}s^{12}r^{24}c^{32}q_0^{84}X_0^{40}Y_0^{52}R_0^4k_0^{12}.$$

Selecting h^{12} in the last factor and z^4 in R_0^4 gives (27) uniquely. The cubic calculation uses $Y_0 = 2vq_0^3$, $k_0 = hX_0Y_0$, and the corresponding leading form with coefficient $2^{24}3^{14}$ and q_0^4. Thus no possible cancellation involving one of the other five fixed ports can affect the nonzero witness.

Without the center, Z has degree 35 and b is the unique dominant radicand. The leading term is b_0^4, giving degrees 456 and 488. The finite auditor checks the complete norm identity, these cancellation cuts, the leading certificates, all declared variables, and the actual graph counts.

The principal graph has 24 additions, 22 subtractions, and 83 multiplications, totaling 129. The centered cubic graph has $24 + 21 + 78 = 123$. The simpler center $3QV$ is sufficient for the exact degree reduction. Subtracting another multiple of V would refine an approximation that is not responsible for the highest remaining degree.

7 A direct quartic in 478 integer variables

The preceding compact equations optimize the number of integer unknowns. There is also a concrete improvement at degree four using the separate direct C222 proof-certificate source. Let $R_0, \ldots, R_{304}$ be the literal integral quadratic polynomials in `direct/equation.json`, in its specified coordinate order, and set

$$\boxed{P_{478}(z_0, \ldots, z_{477}) = \sum_{j=0}^{304} R_j(z)^2.} \tag{29}$$

The companion `expanded_polynomial.json` contains every collected coefficient. A term $[c, [i_1, \ldots, i_d]]$ denotes $cz_{i_1} \cdots z_{i_d}$, with an empty index list denoting the constant monomial. Every coefficient is a literal integer, and every variable ranges independently over $\mathbb{Z}$.

Theorem 7.1. *The polynomial (29) has exact degree four, 478 occurring variables, 5,484 collected monomials and coefficient height 511,326. The ordinary proofs below, composed with the explicitly retained direct C222/refutation calibration, establish*

$$\mathrm{ZFC} \vdash \left[\neg\exists z \in \mathbb{Z}^{478}\ P_{478}(z) = 0\right] \iff \mathrm{Con}(\mathrm{ZFC}).$$

The matched predecessor is the derived-label square-shell endpoint in the incoming powered-saturation/factorial-carrier archive at the comparison snapshot. Its metrics, and the new metrics, are:

	Variables	Quadratic rows	Monomials	Height
Matched predecessor	517	310	6,774	3,904,267
New direct equation	478	305	5,484	511,326

This improves all four displayed metrics for that direct branch. Other repository constructions have fewer variables with different degree or coefficient costs. No global minimum is asserted.

7.1 The dependency order behind the saving

The source supplies a proposed factorial F, a horizon $t = t_0 + 16$, a proposed power V, and exact polynomial relations

$$D = r + s + 1, \qquad DF = V + s, \tag{30}$$

$$q = D + o_0 + 1, \qquad q = B + o_1 + 1, \tag{31}$$

$$Bi = C + F^2 + 1, \qquad C \geq 1. \tag{32}$$

Here D and B are proposed binomial outputs. Their eventual meanings are not assumed in these identities. The power interface requests

$$V = (Ft^2 + 1)^t, \qquad N = B^K, \qquad q = N(N + V + 1)^K, \qquad K = 2(2 + t + x + y + F),$$

as well as the powers used for digit and range extraction. Before power recovery, retain $t_0, x \geq 0$, the endpoint and encoded-input signs, and $o_0 \geq 0$. Strengthen the existing factorial guard to $F \geq 1$. It costs the same number of witnesses and holds for every genuine factorial.

Lemma 7.2 (Factorial signs after one power)**.** *Suppose $F \geq 1$, $V > F$, $D = r+s+1$, $DF = V+s$ and $rs \geq 0$. Then $r, s \geq 0$ and $D \geq 1$.*

Proof. Eliminate D to obtain $V = Fr + (F - 1)s + F$. If either member of (r, s) were negative, product nonnegativity would make both nonpositive. The displayed expression would then give $V \leq F$, a contradiction. This includes the cases where one member is zero. □

The base $Ft^2 + 1$ is already at least two. The signed power certificate, whose frame is justified below independently of these signs, therefore recovers $V = (Ft^2+1)^t > F$ first. Lemma 7.2 proves $D \geq 1$, and hence $q = D + o_0 + 1 \geq 2$. Next require only $Bo_1 \geq 0$. The sum $B + o_1 = q - 1 > 0$ forces $B, o_1 \geq 0$; the reciprocal index identity excludes $B = 0$, so $B \geq 1$ before its power is interpreted.

The two calls at index K now orient their shared Pell pair. The first either gives $N = B^K$ or has the exceptional inverse case $B = 1, N = -1$. In both cases its second base $N + V + 1$ is at least two. That second call fixes the orientation and excludes the inverse case. All remaining seed pairs have supporting calls at q or $q + 1$, both at least two. The unchanged addition formula recovers the single derived $t + 1$ pair. Thus all nineteen requested powers have their intended values.

Only now recover the other delayed signs. Both binomial remainder pairs and all ten range remainder pairs have exact sum equations

$$r_j + s_j = q^{\ell_j} - 1 > 0,$$

where the already positive exponent is $t, t+1, x+1$, or $y+1$. For each such pair, $r_j s_j \geq 0$ implies both coordinates are nonnegative. This replaces fourteen pairs of scalar predicates by fourteen product predicates. No binomial, factorial or CRT interpretation is used to restore a sign that its own interpretation needs.

For unrestricted integer witnesses, use

$$Q(a, b, c) = a^2 + b^2 + c(c + 1).$$

It represents exactly the nonnegative integers. Nonnegativity is immediate, and surjectivity follows from the three-square theorem applied to $4n + 1$: exactly one of the three squares is odd. Consequently $rs = Q(a, b, c)$ is one quadratic row expressing $rs \geq 0$. The ordinary positive-sum/product observation is elementary and already used in earlier repository work; the new saving is its dependency-ordered application to this stronger source, enabled by the earlier factorial bootstrap.

7.2 One guard for orientation and a nonzero shared ordinate

Let A be the main Pell base, and write the shared auxiliary ordinate as

$$V_* = \left(\prod_i Y_i^2\right)(v+1).$$

The retained seed slacks give $Y_i \geq k_i \geq 1$ before any power is interpreted. Separate guards on the frame orientation and on v can be replaced by two quadratic rows involving four fresh coordinates:

$$u = (v+1)^2, \qquad (A-1)u - 1 = \alpha^2 + \beta^2 + \gamma(\gamma+1). \tag{33}$$

Since the right side is nonnegative, $(A-1)u \geq 1$. Thus $A \geq 2$, $u \geq 1$ and $v \neq -1$, so $V_* \neq 0$. The sign of V_* is immaterial. Conversely, canonical positive Pell witnesses have $A \geq 2, v \geq 0$; the left side in the second equation is nonnegative and admits the required three-square witnesses. This joins two useful conclusions in one predicate while keeping the final equation quartic.

The unchanged auxiliary norm gives

$$U^2 - (A^2-1)V_*^2 = 1, \qquad |U| \geq A.$$

The comparison base b satisfies

$$b = 1 + 4V_*(q_4+1), \qquad b \equiv A \pmod{U}.$$

We permit q_4 to have either sign. The first equation gives $b \equiv 1 \pmod 4$; the second excludes $b = 1$ because $0 < A - 1 < |U|$. Therefore $|b| \geq 3$.

The signed comparison-base index theorem is inherited from the incoming quadratic-frames package, and its complete proofs are retained under `direct/source/`. For completeness, the mechanism is as follows. Classify absolute Pell solutions at A and signed solutions at b; polynomial dependence in the base transfers their congruences. Squared divisibility $Y_i^2 \mid V_*$ gives $Y_i \mid h$ for the common auxiliary index h. The abscissa residue identity gives the comparison index congruent to either sign of the main index modulo $2h$. The base-one ordinate congruence gives either sign of k_i modulo $4Y_i$. Hence the main index is congruent to $\pm k_i$ modulo $2Y_i$. Both lie in $[1, Y_i]$, so they agree; the possible common endpoint also gives equality. Thus

$$|X_i| = \chi_A(k_i), \qquad Y_i = \psi_A(k_i).$$

This argument allows both signs of V_*, U, b and the comparison coordinates. It never requires the removed private signs to be pointwise positive.

7.3 A frame that uses proved signs efficiently

The original power frame squared every atom. The new frame uses

$$W = 2 + \sum_{a \text{ already proved } \geq 0} a + \sum_{a \text{ of undetermined sign}} a^2. \tag{34}$$

There are 41 distinct affine atoms: 18 enter linearly and 23 quadratically. Their signs are certified by 163 ordered polynomial deductions from 31 retained scalar predicates. The frame-definition row and every power interpretation are excluded from those deductions. For every integer a, $a^2 \geq |a|$; hence $W \geq 2$ and $W > |a|$ for every base, exponent and proposed output in the finite power interface.

Writing $Z = Z_{\text{minus1}} + 1$, the unchanged frame norm is

$$A^2 - W^3(W+2)Z^2 = 1.$$

Use $\chi_a(j)+\psi_a(j)\sqrt{a^2-1}=(a+\sqrt{a^2-1})^j$. Equation (33) gives $A\geq 2$, so $Z\neq 0$ even though its sign is unrestricted. Pell classification at base $W+1$ and the recurrence $\psi_{W+1}(h)\equiv h \pmod W$ yield $A=\chi_{W+1}(h)$ with $h\geq W$. Thus

$$A\geq (W+1)^W, \qquad T_n=2An-n^2-1>W^W+W \quad (1\leq n<W).$$

For example $T_n\geq 2A-2$, and the binomial expansion gives $(W+1)^W\geq 2W^W$. These bounds hold before any proposed output is interpreted or given a sign.

At an indexed pair, the power row

$$X+(n-A)Y=m+T_nw$$

gives $m\equiv n^k \pmod{T_n}$ for positive X, and $mn^k\equiv -1 \pmod{T_n}$ for negative X. The frame bounds give $|m-n^k|<T_n$ and $|mn^k+1|<T_n$. A base $n\geq 2$ excludes the inverse case and gives $m=n^k$ exactly, with no prior sign assumption on m. This licenses the initial factorial call and the dependency order used above.

7.4 Both directions and the finite equation

Given a zero of the new system, the preceding arguments recover all powers and all semantic front signs. The inherited self-bounding factorial, exact digit, reciprocal prime-power CRT, derived-label bound and pairwise-coprime Horn-flow proofs then recover a finite C222 refutation. Their full ordinary proof texts are included under `direct/source/`. The endpoint composition is `KEndpoint.c222_iff_zf_refutation` followed by `ConstructibleChoice.zf_refutation_iff_zfc_refutation`.

Conversely, start with the inherited arithmetic front supplied by any finite refutation. It has $F=t!\geq 1$ and nonnegative entries in every fused pair. Construct its new W by (34), choose the positive Pell solution at index W for base $W+1$, and choose the main pairs at their required indices. The order of a unit in a finite quadratic ring supplies a common positive auxiliary iterate divisible by the entire product $\prod_i Y_i^2$. CRT supplies a positive comparison base with the required two congruences. Complete all comparison pairs and power quotients. These canonical witnesses have $v\geq 0$, so (33) and every product predicate admit their ordinary three-square witnesses. No consistency assumption is used in this map.

These are equivalences of existential zero assertions. New private comparison coordinates need not solve the old positive-domain auxiliary rows; fresh old witnesses can be reconstructed after recovering the front.

The complete new source has 690 integer coordinates and 515 quadratic rows. The projection certificate records 210 integral monic restorations; two remaining unused coordinates are discarded with restoration zero. Every substitution keeps every residual quadratic. This leaves 478 coordinates and 305 rows. The final explicitly recorded integral translations are bijections. Expanding the sum of squares gives all 5,484 nonzero terms of (29) and proves the stated exact metrics.

The portable builder regenerates all ten normative data files byte for byte. One independent checker reconstructs every guard and frame row; a second checks all source hashes, transformations, projection inverses, translations and collected quartic coefficients. Both pass under normal and optimized Python. Their finite boundary controls support transcription and degeneration checks; they neither decide solvability nor replace the unbounded mathematical proof. This branch uses the direct calculus, without a K215 execution encoding or Jones universal-polynomial parameter.

8 Fixed numeral definitions

The eleven-variable forms retain one complete fixed source34 tuple. It is deliberately separate from the direct quartic's small literal coefficient list. The byte files are actual numerals: a

decompressed little-endian payload $(b_0, \dots, b_{m-1})$ denotes exactly

$$\sum_{i=0}^{m-1} b_i 256^i.$$

Hashes authenticate those bytes and do not stand in for unspecified integers. The six minimal payloads total 615,883 raw bytes.

Port	Bit length	Raw bytes
β	1064921	133116
K_0	1065043	133131
M_C	1064921	133116
M_F	1064922	133116
A_0	1114	140
B_0	666105	83264

For the finite recipe, put $\rho = 2^{121}$ and $\mathcal{B} = \rho^{8801} = 2^{1064921}$. The retained file `wang943_tiles.json` gives the ordered 943 tiles in west–north–east–south order, reconstructed from the included literal 430 transitions. Start is selector 0 and forbidden is selector 1. Let x_{it} be the four labels of tile i, and put

$$\begin{aligned} W_j &= 2^{10+2j}, \qquad (o_0, o_1, o_2, o_3) = (1020, 1000, 980, 960), \\ \mathcal{E} &= \{0, \dots, 942\} \cup \{o_t + 2j : 0 \le t < 4,\ 0 \le j < 10\} \\ &\qquad \cup \{1063, 1064, 1095, 1096, 2195, 3296\}, \\ c_i &= 1 + \sum_{t=0}^{3} \sum_{j=0}^{9} \mathrm{bit}_j(x_{it} + 512) W_{10t+j} + \sum_{j=40}^{43} W_j \quad (0 \le i \le 942). \end{aligned}$$

At the copy exponent o_t+2j put $c_{o_t+2j} = W_{10t+j}$; at $1095, 1096, 2195, 3296$ put $W_{40}, W_{41}, W_{42}, W_{43}$ respectively. The dummy coefficients at $1063, 1064$ are zero. Then

$$\begin{aligned} \mu &= 1022 + \sum_{j=0}^{43} W_j, \\ Z_p &= (2^{119} - 1) \sum_{e=8792}^{8798} \rho^e + (2^{27} - 1)\rho^{8799}, \\ D_C &= \rho^{40} + \rho^{41} + \rho^{1100} + \rho^{2200} + \sum_{e \in \mathcal{E}} c_e (\rho^{4393-e} + \rho^{5495-e}), \\ M_C &= \mathcal{B} - 1 - \sum_{e \in \mathcal{E},\ e \ne 1} \rho^e - 2\rho^{1063}, \\ M_{F,n} &= 4 + \mu(\rho^{4393} + \rho^{5495}) + \sum_{j=0}^{9} (\rho^{1000+2j} + \rho^{1021+2j}) \\ &\qquad + \rho^{2195} + \rho^{3296} + Z_p. \end{aligned}$$

The ports are precisely

$$\beta = \mathcal{B} - 1, \quad K_0 = D_C + \mathcal{B}\rho, \quad M_C, \quad M_F = M_{F,n} + \mathcal{B} - 1, \quad A_0 = 2^{1113}, \quad B_0 = 2^{666104}.$$

The authentic tuple has an ordinary compiler adapter and canonical rank controller, retained in the source material. Matching only these residues or substituting a different tuple's mask would not justify the consistency interpretation.

9 Artifact design, reproduction, and verification boundaries

The archive contains the full manuscript source and PDF, the four literal eleven-variable graphs, the direct residual and expanded polynomial files, deterministic builders, independent checkers, fixed numeral payloads, and scoped inherited proof texts. The top-level `README.md` gives exact commands and the final file map. No network lookup is required to run the new finite checks.

Small-variable graph format. Nodes are topologically ordered integer, coefficient, or variable leaves, followed by binary `add`, `sub`, or `mul` nodes referring to earlier node indices. The output index identifies one polynomial. In particular, register names do not add variables. A reader can reconstruct the expression without executing the builder or trusting its operation count.

Direct sparse format. Each residual is an explicit quadratic integer polynomial. The full quartic is its collected sum of squares, stored separately and re-expanded independently. Every coefficient and variable occurrence is supplied; the final polynomial is not a request to call a proof predicate, a binomial oracle, or a universal interpreter. Arithmetic graphs used during its derivation have already been expanded into ordinary polynomial rows.

What the checks establish. Exact algebra checks transcription, the 35-term norm identity, degree cancellations, literal coefficients, and the direct sum of squares. Graph checks establish variable closure, liveness and counts. Source and coefficient hashes establish byte identity. Reconstruction checks reproduce the supplied new artifacts in a separate output directory. Selected deliberate corruptions must be rejected. Optimized-Python runs check that acceptance does not depend on disabled `assert` statements. The machine-readable receipts state their actual scopes.

These are useful finite certificates. They do not by themselves establish universal Pell classification, supported-modulus interpolation, the native compiler, the C222 calibration, or consistency. Those are mathematical dependencies and ordinary proofs, identified separately. No search for a ZFC contradiction, nor a bounded failure to find an integer solution, is used as evidence of (1).

10 Further research questions

1. **Smaller mixed carriers.** The current mixed chart needs $M^5 \mid A_*$ to recover $M \mid v - q^2$ by a least-valuation argument. The H part only needs first-power divisibility. A carrier tailored to the supported strong modulus might reduce the M^5 cost, but its completeness must work at every prime dividing the chosen strong ordinate, not just at generic primes.

2. **An eleven-integer equation below degree** 416**.** The leading obstruction is the mixed term $16b_0^2e_0^2$. Refining the auxiliary center alone does not remove it. Reducing the budget/guard degree, or proving a different fusion between the guard and a Pell condition, is a more consequential target than another correction to the same center.

3. **Ten integers without the general reduction.** The guard parameter s and strong multiplier r are still independent. Identifying them naively imposes simultaneous Pell conditions with a common ordinate; ordinary Pell existence does not prove that those simultaneous conditions have a solution. A valid elimination must solve that completeness problem explicitly.

4. **Further direct sign sharing.** The 478-variable quartic uses exact source identities to recover individual signs from shared products. Further savings require a proved positive sum or a similarly noncircular certificate. The number of removed guards is not the final

variable saving, since a product can retain coordinates that affine projection previously removed.

5. **Lower coefficients in the direct route.** The new frame lowers height substantially on its matched branch, but more variables can still buy smaller coefficients. An explicit Pareto search should measure collected terms and height after projection, not just gate counts before projection.

6. **Formalize one complete bridge.** The most valuable formal target is the ordinary equivalence between one fully emitted direct polynomial and finite C222 refutability, connected to the already identified endpoints. Local norm identities alone would not close that logical gap.

Attribution and intended integration

The inherited proof and code notices are retained. New prose, circuits, and checks were prepared with AI assistance for incorporation into Vladimir Reshetnikov's ProveIt repository. The work builds on the specific incoming and repository results identified above; classical Pell arithmetic, CRT, norm identities, and intersective polynomials are credited as inherited tools. No repository branch was modified or submission published by producing this archive.

References

[1] Vladimir Reshetnikov, *ProveIt*, source snapshot `45b399ec51df becb891dd51d8a9e4b6b7c361ac6`, 24 October 2026. https://github.com/VladimirReshetnikov/ProveIt. The relevant directories are `Computability/DiophantineConsistency`, `Computability/HilbertTenthProblem`, and `docs/incoming`.

[2] ProveIt incoming research packages, *Coprime Fusion and Low-Degree Diophantine Projections*; *Powered Saturation and Factorial Carriers, Round 14*; *Weaker Congruences and Even Pell Indices*; and *Quadratic Frames*, 24 October 2026. Exact archive names, source bytes, and dependency roles are recorded in this release's provenance ledgers. https://github.com/VladimirReshetnikov/ProveIt/tree/45b399ec51dfbecb891dd51d8a9e4b6b7c361ac6/docs/incoming.

[3] A. M. Hyde, P. D. Lee, and B. K. Spearman, *Polynomials* $(x^3 - n)(x^2 + 3)$ *solvable modulo any integer*, American Mathematical Monthly **121**(4) (2014), 355–358. https://doi.org/10.4169/amer.math.monthly.121.04.355.

[4] Zhi-Wei Sun, *Ten unknowns for Hilbert's tenth problem over the integers*, arXiv:2609.01594v2, 2 September 2026. https://arxiv.org/abs/2609.01594. The present eleven-variable construction does not instantiate that general ten-variable reduction.